IMCCRT-2026

# AI-Based Job Resume Screening

## ENHANCING RECRUITMENT THROUGH INTELLIGENT AUTOMATION

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# Abstract

AI-Based Job Resume Screening is an intelligent system designed to automate and improve the recruitment process by using Artificial Intelligence (AI) and Natural Language Processing (NLP) techniques. The system analyzes resumes submitted by candidates and compares their qualifications, skills, education, experience, and keywords with the requirements specified in a job description. Based on the matching results, it generates a suitability score and ranks candidates according to their relevance to the job role. This reduces the time and effort required for manual resume screening and helps recruiters manage a large number of applications efficiently. The system can also identify skill gaps and highlight the most relevant information in each resume. By automating repetitive recruitment tasks, the proposed system aims to improve screening speed, consistency, and efficiency. However, human supervision is essential to reduce bias, protect candidate privacy, and ensure fair hiring decisions. Overall, the AI-Based Job Resume Screening system provides an effective solution for streamlining the initial recruitment process and supporting better candidate selection.

# Problem Statement

1. In the traditional recruitment process, recruiters manually check a large number of resumes to find suitable candidates for a job.

2. It is also difficult to identify the right candidates based on their skills, qualifications, and experience.

3. This system helps reduce manual work, save time, and make the recruitment process faster and more efficient.

# Objectives

1. To automate the resume screening process using Artificial Intelligence.
2. To identify suitable candidates based on their skills, qualifications, and experience.
3. To improve the speed and efficiency of the recruitment process.
4. To help recruiters select the right candidates more easily.

# Literature Survey

### Study 1: Rojas Galeano (2022)

The researchers studied how Artificial Intelligence and Natural Language Processing (NLP) can be used to match resumes with job descriptions. Their study explains how AI helps extract skills and qualifications from resumes and identify suitable candidates.

### Study 2: Modak et al. (2024)

The researchers reviewed machine learning and NLP techniques for resume analysis and job description matching. Their study explains how these techniques help identify important skills, qualifications, and experience from resumes.

# Research Gap

Existing systems may have difficulty understanding different resume formats, identifying skills expressed in different words, and avoiding biased results. Therefore, the proposed AI-Based Job Resume Screening system aims to analyze resumes, match skills with job descriptions, and rank candidates according to their suitability while supporting human decision-making, and then we can add multiple resumes for screening and get the result in the bar graph.

# Proposed System

The proposed system uses Artificial Intelligence (AI) and Natural Language Processing (NLP) to automatically screen resumes and identify suitable candidates for a job. This system reduces manual work, saves time, and helps recruiters shortlist candidates more efficiently.

## How It Works

**Resume Upload:** Allows candidates to upload their resumes.

**Text Extraction:** Extracts important information from resumes.

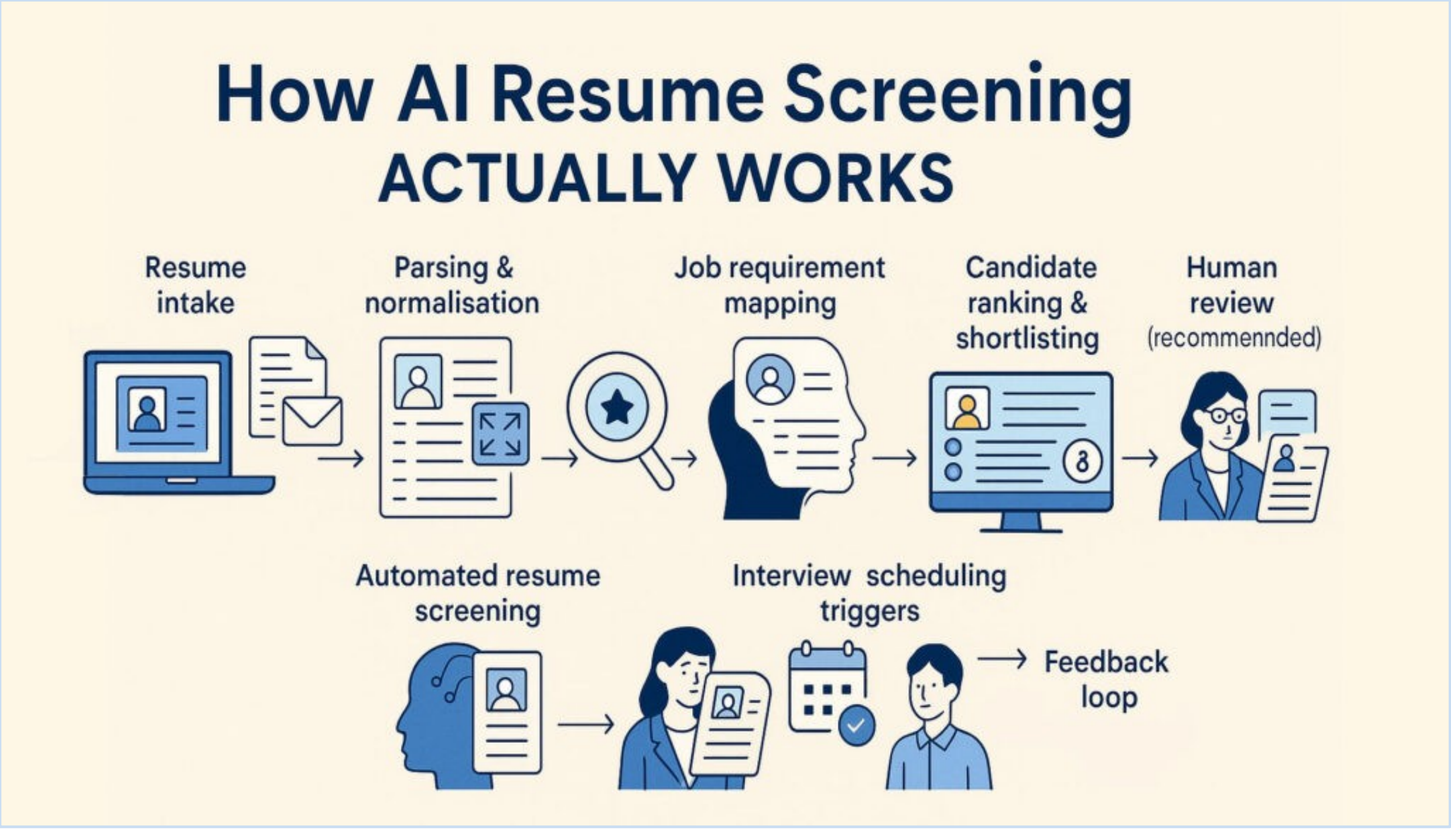
**Skill Matching:** Matches candidate skills with job requirements.

**Resume Ranking:** Ranks candidates based on their matching scores.

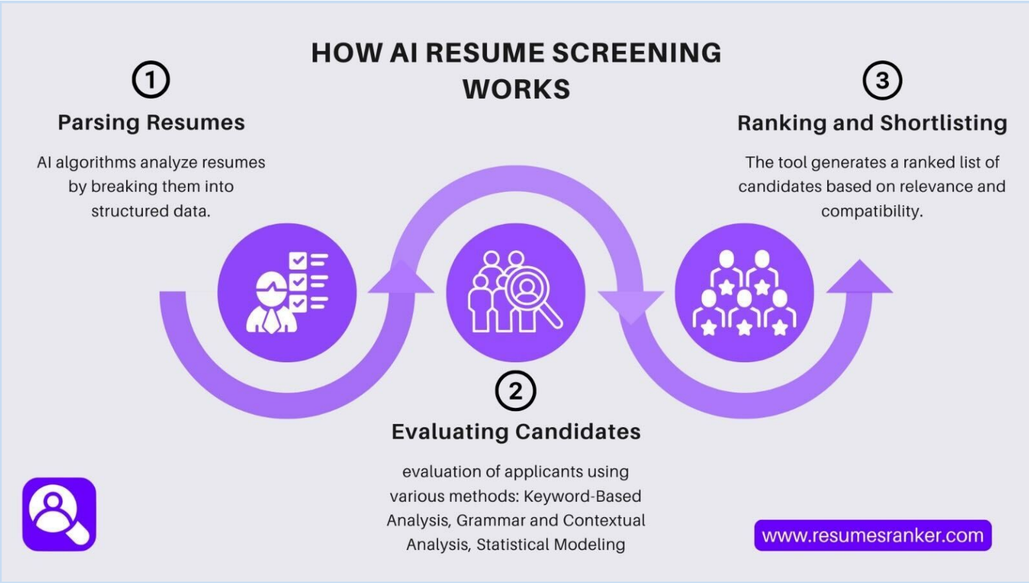
**Candidate Shortlisting:** Helps recruiters identify suitable candidates.

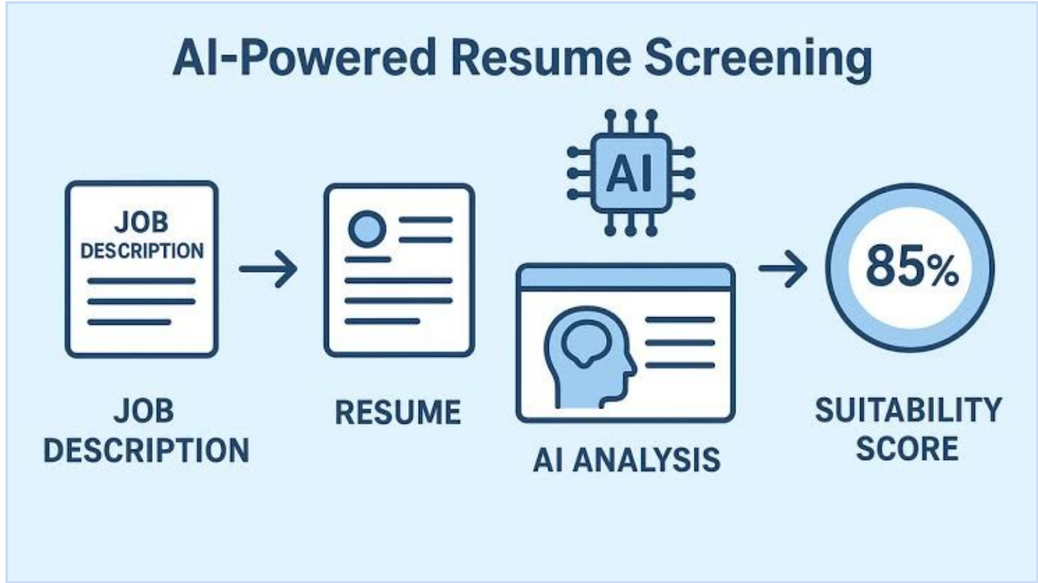
**Result Display:** Shows matching scores and candidate details.

# System Architecture



# Architecture



# Methodology & Technology Used

## Methodology

1. Upload resumes and enter the job description.
2. Extract text from resumes.
3. Identify skills, education, and experience using NLP.
4. Compare resumes with job requirements.
5. Calculate matching scores and rank candidates.
6. Display results for recruiter review.

## Technology Used

- **Python:** Main programming language.
- **HTML and CSS:** User interface design.
- **Flask:** Backend development.
- **NLP:** Extracts and analyzes resume information.
- **TF-IDF and Cosine Similarity:** Matches resumes with job descriptions.
- **PyMuPDF:** Extracts text from PDF resumes.
- **MySQL / SQLite:** Stores candidate details and results.

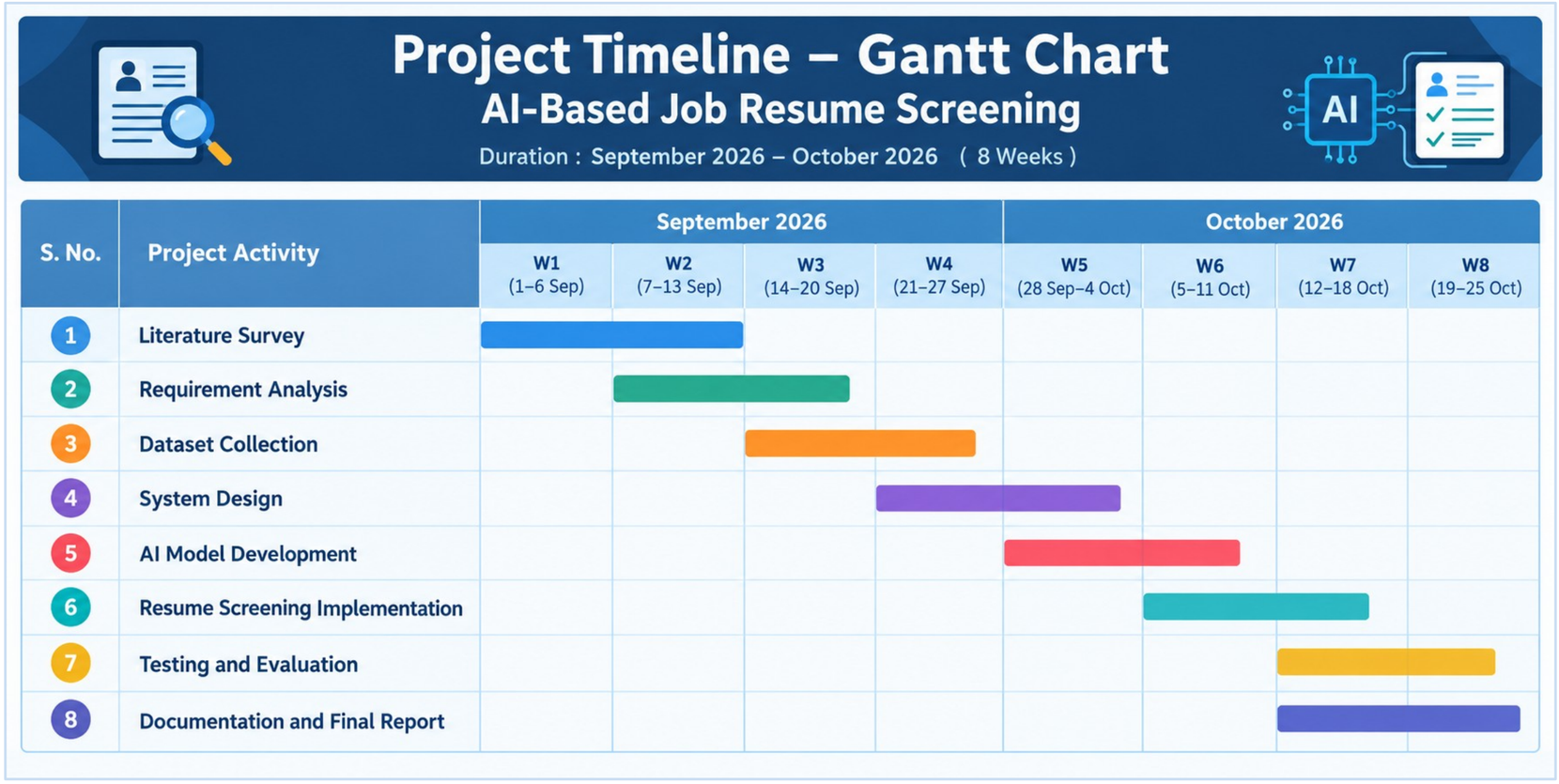
# Future Enhancements – AI-Based Job Resume Screening

- **Advanced AI:** Use advanced AI models to improve resume analysis.
- **Multiple Formats:** Support more resume formats such as PDF, DOCX, and scanned images.
- **Skill Gap Analysis:** Identify missing skills in candidate resumes.
- **Job Recommendations:** Suggest suitable jobs based on candidate skills.
- **Bias Reduction:** Improve fairness in candidate screening.
- **Real-Time Screening:** Process large numbers of resumes quickly.
- **Automated Notifications:** Send interview updates to shortlisted candidates.

# Conclusion

The AI-Based Job Resume Screening system helps the Human Resources (HR) department simplify the recruitment process by automatically analyzing resumes and identifying suitable candidates. It matches candidates' skills, qualifications, and experience with job requirements, saving time and reducing manual work. This system helps HR professionals shortlist candidates more efficiently and manage large numbers of applications. However, the final hiring decision should be made by HR professionals to ensure fair and accurate candidate selection.

# Gantt Chart


Project Timeline – Gantt Chart
AI-Based Job Resume Screening
Duration : September 2026 – October 2026 ( 8 Weeks )
AI
S. No.
Project Activity
September 2026
October 2026
W1 (1–6 Sep)
W2 (7–13 Sep)
W3 (14–20 Sep)
W4 (21–27 Sep)
W5 (28 Sep–4 Oct)
W6 (5–11 Oct)
W7 (12–18 Oct)
W8 (19–25 Oct)
1 Literature Survey
2 Requirement Analysis
3 Dataset Collection
4 System Design
5 AI Model Development
6 Resume Screening Implementation
7 Testing and Evaluation
8 Documentation and Final Report

# Thank You